INFORMS 2026 Data Mining Society Data Challenge

Outage Severity Index (OSI): Methodology and Data Quality Notes

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This document details the OSI construction, component formulas, and data quality notes. The OSI formula overview is in `problem_description.pdf`. Variable descriptions for all columns are in `variable_descriptions.pdf`.

OSI Component Formulas

The OSI is built from four hourly variables derived from `outageCount` and `customersTracked`. All four (P_t, N_t, D_t, R_t) are provided in both the train and test files for March 11–13. They are withheld (NaN) for March 14–19 in the test file.

Var	Type	Formula	Captures
`P_t`	State	`outageCount(t) / customersTracked`	Current outage fraction (0–1)
`N_t`	Flow	`max(0, ΔoutageCount) / customersTracked`	New outage growth this hour
`D_t`	State	`rolling_mean(P_t, 6 hours)`	6-hour sustained persistence
`R_t`	Flow	`max(0, −ΔoutageCount) / customersTracked`	Active restoration rate

Composite OSI formula:

$$\mathrm{OSI}_t = 0.40\,P_t + 0.35\,N_t + 0.25\,D_t - 0.10\,R_t \qquad (\text{clipped} \geq 0)$$

Restoration actively subtracts from severity; a county where crews are actively restoring power scores lower than one with the same outage level but no restoration activity. Typical range: 0–0.60; theoretical maximum ≈ 0.90.

Reconstructing OSI for the Test File

The test file provides all four OSI components (P_t, N_t, D_t, R_t) for the March 11–13 observed window. OSI and its lags can be reconstructed from these components using the formula above for those rows.

Note that any reconstruction must respect the temporal causality rule: when making a prediction at origin hour t, OSI lag features may only be derived from rows at time t or earlier within the same county. Outage variables for March 14–19 are withheld (NaN) in the test file and must not be imputed or filled; they are the prediction target window.

Train / Test Split

80/20 county-level split, stratified by state × severity level. Severity combines peak OSI rank and average outage duration rank within state. Random seed: 42.

State	Train	Test	Total
IN	72	20	92
OH	70	18	88
PA	52	15	67
WV	45	10	55
Total	**239**	**63**	**302**

Each county appears in exactly one of train or test; there is no overlap. The `severity_tier` column (0–4) is provided in `DM_Train.csv` for stratified cross-validation only. It is absent from `DM_Test.csv` and must not be used as a model input feature. In the test file, outage variables are present only for March 11–13 (the first 72 hours); they are NaN for March 14–19. Weather variables are present for all 216 hours in both files.

Data Quality Notes

- **outage_pct cap:** `outage_pct` and P_t are capped at 1.0 (100%) for two counties whose `outageCount` exceeded `customersTracked` due to service-territory boundary mismatch.
- **customersTracked variation:** Varies slightly across hours for 204 of 239 train counties due to multi-utility reporting timing. The variation is small (median within-county range: 0.05%) and all outage fractions are internally consistent using the row-level denominator. No correction is applied.
- **Pre-event outages:** The March 11–12 pre-event window is not a clean baseline. Lingering outages from a prior February event are present in 236 of 239 train counties (one county, FIPS 18111, shows 100% outage rate in the pre-event window; this is real prior-state signal, not an error). Do not zero-fill or exclude pre-event rows; they carry valid lag signal for `osi_lag24h` and `osi_lag48h` at storm onset.
- **Two-storm structure:** Two distinct wind events occurred within the March 13–19 window. OSI trajectories for many counties show two humps. The `osi_lag24h` and `osi_lag48h` features are specifically useful for distinguishing first-wave vs second-wave dynamics.
- **NaN in targets:** OSI target columns are NaN for the last k rows per county per horizon (last 1 row for t+1h, last 6 for t+6h, last 24 for t+24h, last 48 for t+48h). Drop these NaN rows during training. Leave them as NaN in your submission; they are not scored.